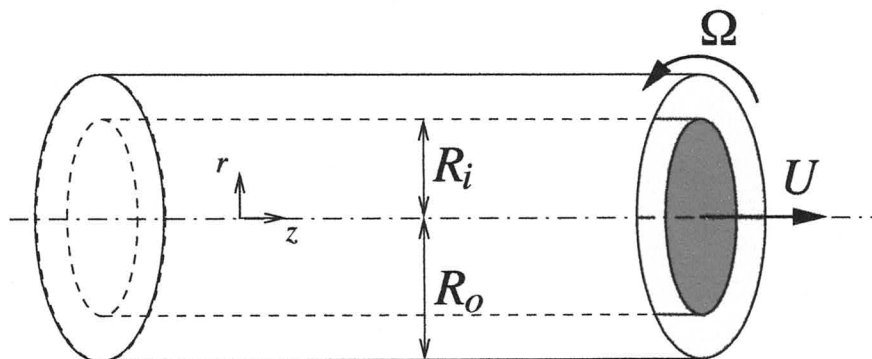


A liquid of density ρ , viscosity μ , specific heat c , and thermal conductivity k fills the annular gap formed between two concentric cylinders of radii R_i and R_o , respectively. The inner cylinder moves with axial velocity U while the outer cylinder rotates around its axis with angular velocity Ω , inducing in the liquid an axially symmetric motion described in cylindrical coordinates by the velocity components $v_r(r)$, $v_z(r)$, and $v_\theta(r)$ with associated pressure distribution $p(r)$.

1. Obtain the radial velocity $v_r(r)$.
2. Calculate the axial velocity $v_z(r)$.
3. Determine the azimuthal velocity $v_\theta(r)$.
4. Compute the pressure distribution, giving the result in the form $p(r) - p_i$, where p_i is the known pressure on the surface of the inner cylinder.
5. Give the trajectory of the fluid particle located initially at $r = r_o$, $\theta = \theta_o$, and $z = z_o$. Show schematically the resulting path line.
6. Obtain the temperature distribution in the liquid $T(r)$ knowing that the temperatures of the inner and outer cylinders are kept at constant values T_i and T_o , respectively.



$$1) \frac{\partial}{\partial r}(rv_r) = 0, \quad rv_r = \text{CONSTANT} = 0 \Rightarrow \boxed{v_r = 0} \quad \begin{matrix} \text{at } R=R_i, R=R_o \\ v_r = 0 \end{matrix} \quad 2) \quad 0 = \frac{1}{r} \frac{\partial}{\partial r} \left(r \frac{\partial v_z}{\partial r} \right) \begin{cases} r=R_i: v_z = U \\ r=R_o: v_z = 0 \end{cases} \Rightarrow \boxed{v_z = U \frac{\ln(R_o/r)}{\ln(R_o/R_i)}}$$

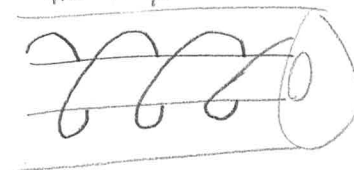
$$3) \quad 0 = \mu \left[\frac{\partial}{\partial r} \left(\frac{1}{r} \frac{\partial}{\partial r} (r v_\theta) \right) \right] \Rightarrow v_\theta = C_1 r + \frac{C_2}{r} \quad \begin{cases} r=R_i: v_\theta = 0 \\ r=R_o: v_\theta = \Omega R_o \end{cases} \Rightarrow \boxed{v_\theta = \frac{\Omega R_o^2}{R_o^2 - R_i^2} \left(\frac{r^2 - R_i^2}{r} \right)}$$

$$4) \quad -\frac{g v_\theta^2}{r} = -\frac{\partial p}{\partial r}, \quad r=R_i, \quad p=p_i \Rightarrow \boxed{\frac{p - p_i}{\rho \Omega^2 R_o^2} = \frac{R_i R_o}{R_o^2 - R_i^2} \left[\frac{1}{2} \left(\frac{r}{R_i} \right)^2 - \frac{1}{2} \left(\frac{R_i}{r} \right)^2 - 2 \ln \left(\frac{r}{R_i} \right) \right]}$$

$$5) \quad \frac{dr}{dt} = 0, \quad r(0) = r_o \Rightarrow \boxed{r = r_o} \quad \frac{d\theta}{dt} = \frac{\Omega R_o^2}{R_o^2 - R_i^2} \left(\frac{r_o^2 - R_i^2}{r_o} \right), \quad \theta(0) = \theta_o \Rightarrow \boxed{\theta - \theta_o = \frac{\Omega R_o^2}{R_o^2 - R_i^2} \left(\frac{r_o^2 - R_i^2}{r_o} \right) t}$$

$$\frac{dz}{dt} = U \frac{\ln(R_o/r)}{\ln(R_o/R_i)}, \quad z(0) = z_o \Rightarrow \boxed{z - z_o = U \frac{\ln(R_o/r_o)}{\ln(R_o/R_i)} t}$$

THE FLUID PARTICLE
FOLLOWS A SPIRAL
PATH



$$6) \quad 0 = k \frac{1}{r} \frac{\partial}{\partial r} \left(r \frac{\partial T}{\partial r} \right) + \mu \left\{ \left[r \frac{\partial}{\partial r} \left(\frac{v_\theta}{r} \right) \right]^2 + \left(\frac{\partial v_z}{\partial r} \right)^2 \right\} \begin{cases} T(R_i) = T_i \\ T(R_o) = T_o \end{cases} \Rightarrow \begin{aligned} T - T_i &= \frac{\mu}{4k} \left(\frac{\Omega R_o^2 R_o^2}{R_o^2 - R_i^2} \right)^2 \left(\frac{1}{R_i^2} - \frac{1}{r^2} \right) + \frac{1}{2} \frac{\mu}{k} U^2 \ln^2 \left(\frac{R_o}{R_i} \right) (\ln^2 R_i - \ln^2 r) \\ &+ (T_o - T_i) \frac{\ln(r/R_i)}{\ln(R_o/R_i)} \left[\frac{\mu}{4k} \left(\frac{\Omega R_o^2 R_o^2}{R_o^2 - R_i^2} \right)^2 \left(\frac{1}{R_i^2} - \frac{1}{R_o^2} \right) + \frac{1}{2} \frac{\mu}{k} U^2 \ln^2 \left(\frac{R_o}{R_i} \right) (\ln^2 R_i - \ln^2 R_o) \right] \end{aligned}$$